Question			Answer	Marks	AO	Guidance
7	(a)		$(k =)\ 1.5$	B1 [1]	3.3	• Allow embedded in equation e.g. $P_{t+1} = \frac{3}{2}P_t$
7	(b)		162	B1 [1]	3.4	• cao
7	(c)	(i)	$48e^{0.4} = 71.6(\approx 72)$ $48e^{0.8} = 106.8(\approx 107)$ Comment comparing *their* values with the data e.g. • Reasonably close • Not exactly right	M1 A1 [2]	3.4 3.5a	• Both values – might not round answers • Allow $106.8 \approx 108$ • Condone incorrect values if $48e^{0.4}$ and $48e^{0.8}$ seen • Must have correct values • e.g. 71.6 is close to 72 and 106.8 is close to 108 • Ignore attempts to compare the two models • Condone e.g. 'close for $t = 1$' **and** 'not close for $t = 2$'
7	(c)	(ii)	$\frac{dQ}{dt} = 19.2e^{0.4t}$ $\frac{dQ}{dt} = 0.4Q$ so the rate of change is proportional to the population	M1 A1 [2]	1.1 2.2a	• Differentiation – might not simplify e.g. $\frac{dQ}{dt} = 19.2e^{0.4t}$ or $\frac{dQ}{dt} = \frac{96}{5}e^{0.4t}$ • Implied by $\frac{dQ}{dt} = ke^{0.4t}$ • Condone poor/no notation • Showing the rate of change is proportional to population e.g. $\frac{dQ}{dt} = 0.4 \times 48e^{0.4t}$ **and** $Q = 48e^{0.4t}$ soi or $\frac{\frac{dQ}{dt}}{48e^{0.4t}} = 0.4$ • 0.4 must be exact and not rounded e.g. from numerical trials • Must see $\frac{dQ}{dt}$ or e.g. Q'

Question			Answer	Marks	AO	Guidance
7	(c)	(iii)	Method to find multiplier for Q At time $t+1$, Q$= 48e^{0.4(t+1)}$ $= 48e^{0.4t} \times e^{0.4}$	**M1**	**3.1b**	• Use of exponential equation one month later – may award for correct use of equation for Q to find populations for at least 2 consecutive months (for $t = 2$ onwards) For info: $t = 2, Q = 106.8$; $t = 3, Q = 159.4$; $t = 4, Q = 237.7$ to 3 s.f. or better • This **M1** may be implied by finding multiplier of $e^{0.4}$
			$e^{0.4}$ or 1.49 ... (is multiplier for Q)	**B1**	**1.1**	• Finding multiplier for Q • If 'multiplier' omitted, then multiplication must be implied by later work
			Multiplier for sequence P is 1.5. Both sequences are equal at $t = 0$ or 1 So ×1.49 each month will give smaller answers than ×1.5	**A1**	**2.4**	• Completion with **convincing** general explanation **including** consideration of an early value (could be $P > Q$ at $t = 1$ or both sequences start at 48 or at $t = 2$ the values are 107 for Q and 108 for P)
			Alternative 1 for A1 Model P is $48(1.5)^t$ oe $1.49 \ldots < 1.5$ So, $48(1.49 \ldots)^t < 48(1.5)^t$	**A1**		• **Convincing** completion, must see '48' or both equal populations at $t = 0$ (or $t = 1$) • oe e.g. showing $\frac{1.5^t}{e^{0.4t}} > 1$ **and** starting values of 48 soi or writing model for P as $48e^{(\ln 1.5)t} = 48\ e^{0.405\ldots t}$
			Alternative 2 for A1 Model P is $48(1.5)^t$ $\frac{dP}{dt} = 48 \times \ln(1.5) \times 1.5^t = \ln(1.5) \times P$ Rate of population growth for P is greater than rate of population growth for Q as $\ln 1.5 > 0.4$ **and** both equal populations at $t = 0$ (or $t = 1$)	**A1**		• Differentiates may use ln 1.5 or 0.405… • **Convincing** completion

Question			Answer	Marks	AO	Guidance
			Alternative method using inequalities Model for P is 48×1.5^t	**B1**		• B1 for finding correct model for P
			$(48)e^{0.4t} < (48 \times)1.5^t$ oe	**M1**		• Forming inequality for P and Q
			$0.4t < \ln 1.5^t$ $0.4t < t \ln 1.5$ True for $t > 1$ since $\ln 1.5 = 0.405\ldots > 0.4$	**A1**		• **Convincing** completion
				[3]		